TOR (Term Of Reference)

GRAID “ Supreme Raid ”

อุปกรณ์ทำแบคอัพข้อมูลด้วยวิธีการทำเรท(Raid) และปลดล๊อคคอขวดบนเครื่องแม่ข่าย

อุปกรณ์การทำเรท Raid ที่สามาถนำ Hard Disk ตั้งแต่ 2 ลูกขึ้นไปมาเชื่อมเข้าด้วยกันเพื่อให้มองเห็นเป็นก้อนเดียวกัน โดยประโยชน์หลักคือการป้องกันการเสียหายของข้อมูล และเป็นการเพิ่มประสิทธิภาพในการอ่าน/เขียนของข้อมูล บนเครื่องแม่ข่าย (Server) หรือ เครื่องเวิร์คสเตชั่น (Workstation)

<u>อุปกรณ์ “ Supreme Raid ” จะต้องมีคุณสมบัติเบื้องต้น ดังนี้</u>

1. รองรับการทำงานแบบไม่มีสายเชื่อมต่อใดๆจากดีสมายังอุปกรณ์ และมีการทำงานแบบ Hybrid RAID (Hardware + Software) ได้
2. รองรับการทำ RAID แบบ 0, 1, 5, 6, 10, NVMe-oF ได้เป็นอย่างน้อย
3. รองรับการทำRaid ได้สูงสุด 8กลุ่ม ในอุปกรณ์เพียงตัวเดียว (Linux 8 / Windows 4)
4. รองรับการเชื่อมต่ออุปกรณ์บันทึกข้อมูล ลักษณะ SSD Disk, NVMe ได้เป็นอย่างน้อย
5. อุปกรณ์ต้องรองรับจำนวนSSD Disk ได้ สูงสุด 32ลูก ในการ์ดใบเดียว
6. รองรับ การเชื่อมต่อแบบ PCIe Gen 3, Gen 4 ได้เป็นอย่างน้อย
7. รองรับ OS ได้หลายระบบ เช่น Windows Server, Linux ได้เป็นอย่างน้อย
8. ต้องมีซอร์ฟแวร์ที่สามารถตรวจสอบและบริหารจัดการได้ในรูปแบบกราฟฟิค GUI (Graphics User Interface)
9. รองรับการป้องกันจัดเก็บไฟล์แบบกระจายตัว (Distributed Data Journaling Protection)
10. รองรับการทำงานที่มีความเสถียรสูง HA (High Availability) เพื่อความต่อเนื่องในการทำงานให้มีประสิทธิภาพสูงสุด
11. รองรับกลไกการลองใหม่เมื่อเกิดข้อผิดพลาด (Error Retry Mechanism)
12. รองรับการฟื้นฟูระบบแบบไร้รอยต่อ (Non-Disruptive Rebuild)
13. มีการรับประกันอย่างน้อย 3 ปี

SR-1001-FD04/ SR-1001-FD08

SupremeRAID™ SR-1001 Series

Product Specifications

Graid Technology Inc.

Unable to Find a Suitable Data Protection Solution for NVMe SSDs?

Graid Technology is the world first solution that breaks through the performance bottleneck of existing NVMe RAID cards and supports **Gen3/4/5** NVMe, SATA, and SAS compatible hybrid RAID cards. The GPU-based SupremeRAID™ provides faster data read/write protection, effectively offloading the CPU, and optimizing data center setup costs.

SR-1001 Software Specs

Supported RAID levels:
RAID 0, 1, 5, 6, 10, 50*, 60*

Max Physical Drives: 32

Max Drive Groups:
Linux: 8 / Windows: 4

OS Support:
AlmaLinux 8
CentOS 7 / 8
Debian 11
openSUSE Leap 15
Oracle Linux 7 / 8 / 9
SLES 15
RHEL 7 / 8 / 9
Rocky Linux 8
Ubuntu 20.04 / 22.04
Windows Server 2019 / 2022
Windows 11

Max Virtual Drives per Drive Group:
Linux: 1023 / Windows: 8

Max Drive Group Size:
Defined by physical drive size

Supported NVMe SSDs:
Memblaze, PETAiO, Dapustor, Hagiwara, Kingston Technologies, KIOXIA, Micron, Phison, Samsung, Scaleflux, Seagate, Solidigm, Western Digital

Supported Platforms:
AMD, Intel

Supported Virtualization Environments:
KVM, Proxmox VE, Virtuozzo OpenVZ, VMWare Workstation Pro 17, Windows Server Hyper-V

*RAID level support RAID 0/1/5/6/10 by SupremeRAID™ and RAID 50/60 only supported by Linux LVM

SR-1001 Card Spcs

Host Interface:
x16 PCIe Gen 3.0

Max Power Consumption:
30 W

Form Factor:
2.713" H x 6.137" L, Single Slot

Product Weight:
132.6 g

Comprehensive Data Protection

SupremeRAID™ supports RAID 0/1/5/6/10/50/60 and consistensy check, controller failover (HA) and write-hole protection to ensure the secure data protection when abnormal power outage.

World Record Performance

Unprecedented NVMe/NVMeoF performance up to 28M IOPS and 260GB/s throughput with a single SupremeRAID™ card delivers the full value of your server investment.

Highly Scalable Applications

Easily manage 32 direct attached NVMe SSDs, No extra PCIe switch is needed with Software Composable Infrastructure.

User-Friendly Management

Out of the box, into the future: effortless installation, no cabling or motherboard re-layout required, easy for BOM and cost control.

Liberate CPU Resources

Offload your entire RAID computation to SupremeRAID™ to free-up CPU computing resources for AI, HPC and Database applications

Easy to Maitain

SupremeRAID™ doesn't rely on memory caching technology, eliminating the need for battery backup modules.

APICTA Asia Pacific ICT Alliance AWARDS

2023 APICTA Golden Award in BS Category

| 2023. 12 | APICTA, Hong Kong

"SupremeRAID™, a GPU-based NVMe RAID solution, stood out among 256 entries in different categories due to its innovation. Judges were especially impressed with its ability to maximize NVMe SSD performance in AI and high-performance computing applications."

Learn More
Please Contact Sales Rep.

SR-1001-FD08/ SR-1001-FD12

SupremeRAID™ SR-1001 Series

Graid Technology Inc.

Product Specifications

World Record Performance –

Since 2021 Graid named over 10 famous awards in worldwide. Named Golden Award in APICTA which is an organization in the Asia-Pacific region that acknowledges outstanding achievements in ICT.

Don't get left behind, experience how SupremeRAID™ to unleash the bottleneck of NVMe RAID card and maxmize the Ultimate performance of your SSD.
Contact Us Today!

6M	100GB/s	up to 100%	80%	7x
IOPS	Throughput	SSD Performance	Cost Savings	Faster Overall

	SupremeRAID™ SR-1001-FD08	SupremeRAID™ SR-1001-FD04	Main Stream Hardware RAID
4K Random Read	6 M IOPS	5.7 M IOPS	3.9 M IOPS
4K Random Write	500 K IOPS	500 K IOPS	180 K IOPS
1M Sequential Read	100 GB/s	50 GB/s	13.5 GB/s
1M Sequential Write	30 GB/s	15 GB/s	4 GB/s
4K Random Read (Rebuild)	1 M IOPS	1 M IOPS	36 K IOPS
4K Random Write (Rebuild)	350 K IOPS	350 K IOPS	18 K IOPS
CPU Utilization	None	None	None
RAID levels	RAID 0, 1, 5, 6, 10, 50*, 60*	RAID 0, 1, 5, 6, 10, 50*, 60*	RAID 0, 1, 5, 6, 50, 60
Controller Failover	Yes - Active/Passive	Yes - Active/Passive	No
Write-hole protection	Yes (Journal)	Yes (Journal)	Incomplete
NVMeoF Support	Yes	Yes	Incomplete(Only for Target)
Max NVMe SSDs Supported	8	4	4
SSD Form Factor	U.2,U.3,E1.s,E3.s,AIC,M.2	U.2,U.3,E1.s,E3.s,AIC,M.2	U.2, U.3

Testing Enviroment

Linux : Server: Supermicro AS-2125HS-TNR x1; CPU: AMD EPYC 9654 96-Core Processor x2; Memory: Samsung M321R2GA3BB6-CQKVS DDR5 4800 MT/s 16GB x24; NVMe SSD: KIOXIA CM7-R 3.84T KCMY1RUG3T84 x8; RAID Controller: SR-1001 x1; Linux Distro: Ubuntu 22.04.1 LTS; Kernel: 5.15.0-83-generic; Benchmarking tool: fio-3.16; SupremeRAID™ Driver version: 1.5.0-670.g03a5380c.001gcf5e69d8

*by Linux LVM

SupremeRAID™:

Protecting NVMe-based Data From The Edge To The Cloud

Graid Technology Inc. APAC regional headquater is in Taipei, Taiwan. Our leadership is composed of a dedicated team of experts with decades of experience in the SDS, ASIC and storage industries. Graid Technology's extraordinary software plus hardware solution has redefined the value of SSD RAID cards and makes SupremeRAID™ the most powerful and flexible NVMe SSD RAID in the world. To learn more, contact us.

SupremeRAID™ SR-1010

Graid Technology Inc.

Product Specifications

Unable to Find a Suitable Data Protection Solution for NVMe SSDs?
Graid Technology is the world first solution that breaks through the performance bottleneck of NVMe RAID cards and supports **Gen3/4/5** NVMe, SATA, and SAS compatible hybrid RAID cards. The GPU-based SupremeRAID™ provides faster data read/write protection, effectively offloading the CPU, and optimizing data center setup costs.

SR-1010 Software Specs

Supported RAID levels:
RAID 0, 1, 5, 6, 10, 50*, 60*

Max Physical Drives: 32

Max Drive Groups:
Linux: 8 / Windows: 4

OS Support:
AlmaLinux 8
CentOS 7 / 8
Debian 11
openSUSE Leap 15
Oracle Linux 7 / 8 / 9
SLES 15
RHEL 7 / 8 / 9
Rocky Linux 8
Ubuntu 20.04 / 22.04
Windows Server 2019 / 2022
Windows 11

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Supported Platforms:
AMD, Intel

Supported Virtualization Environments:
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SR-1010 Card Specs

Host Interface:
x16 PCIe Gen 4.0

Max Power Consumption:
70 W

Form Factor:
2.713" H x 6.6" L, Dual Slot

Product Weight:
306 g

Comprehensive Data Protection
SupremeRAID™ supports RAID 0/1/5/6/10/50/60 and consistensy check, controller failover (HA) and write-hole protection to ensure the secure data protection when abnormal power outage.

World Record Performance
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Contact Us Today!

28M IOPS

260GB/s Throughput

up to 100% SSD Performance

80% Cost Savings

9x Faster Overall

	SupremeRAID™ SR-1010	High-end HardwareRAID
4K Random Read	28 M IOPS	6.9 M IOPS
4K Random Write	2 M IOPS	651 K IOPS
1M Sequential Read	260 GB/s	28.2 GB/s
1M Sequential Write	100 GB/s	10.4 GB/s
4K Random Read (Rebuild)	5.5 M IOPS	1 M IOPS
4K Random Write (Rebuild)	1.1 M IOPS	548 K IOPS
CPU Utilization	None	None
RAID levels	RAID 0, 1, 5, 6, 10, 50*, 60*	RAID 0, 1, 5, 6, 50, 60
Controller Failover	Yes - Active/Passive	No
Write-hole protection	Yes (Journal)	Incomplete
NVMeoF Support	Yes	Incomplete (Only for Target)
Max NVMe SSDs Supported	32	8
SSD Form Factor	U.2,U.3,E1.s,E3.s,AIC,M.2	U.2, U.3

Testing Environment

Linux : Server: Supermicro AS-2125HS-TNR x1; CPU: AMD EPYC 9654 96-Core Processor x2; Memory: Samsung M321R2GA3BB6-CQKVS DDR5 16GB x24; SSD: Kioxia CM7 KCMY1RUG3T84 x24; RAID Controller: SR-1010 x1; OS: Ubuntu 20.04.4 LTS; Kernel: 5.4.0-155-generic; Benchmarking tool: fio-3.16; SupremeRAID™ Driver version: 1.5.0-rc1-20230804.gcf5e69d8

*by Linux LVM

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